Practical Livelock Analysis in Self-Disabling Unidirectional Ring Protocols

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Abstract

We develop a practical framework for livelock analysis in self-disabling unidirectional ring protocols. Klinkhamer and Ebnenasir established that livelock detection for parameterized rings is Π_1^0-complete via reduction from the periodic domino problem. We observe that lifting the analysis from the transition space to an *equivariant product space*—the space of transition–witness pairs—reveals structure that supports effective verification.

We construct a *product transition graph* (at most $|T|^2$ nodes) that captures all livelocks: every livelock maps into this graph as a witness-closed subgraph. The maximal such subgraph $G^*(T)$ is computable in polynomial time ($O(|T|^8)$ worst case) via monotone fixed-point iteration. When $G^*(T) = \emptyset$, the protocol is *provably livelock-free* for all ring sizes—a sound and complete livelock-freedom verifier.

When $G^*(T) \neq \emptyset$, we apply a backtracking search that backward-propagates each simple cycle through G^* until the chain either closes into a torus (confirming a livelock) or dies (no livelock from that cycle). This two-phase algorithm—polynomial pruning followed by combinatorial verification—produces three outcomes: Free, Livelock, or Inconclusive. Across 4,349 protocols tested (including an adversarial protocol derived from Klinkhamer and Ebnenasir's tiling construction and Kari's 14-tile aperiodic set converted via their SE gadget), the algorithm is conclusive in every case with zero errors.

Python implementation and usage instructions are at `https://github.com/cosmoparadox/mathematical-tools`.

1 Introduction

Self-stabilization [6] guarantees that a distributed system recovers from arbitrary transient faults to a legitimate configuration, regardless of the initial state. A key obstacle to self-stabilization in ring protocols is *livelock*: an infinite execution in which processes continually change state but never reach legitimacy. Determining whether a given protocol admits a livelock—for any ring size—is a fundamental parameterized verification problem.

Background. Klinkhamer and Ebnenasir [16] established that livelock detection for parameterized rings of self-disabling processes is Π_1^0-complete, via a reduction from the periodic domino problem. Their torus characterization—the equivalence between livelocks and doubly-periodic tilings—is foundational and underlies our work. Notably, they also showed that the *synthesis* of self-stabilizing rings is decidable [16], and Farahat's equivariance framework [10] has been used to prove livelock freedom of difficult protocols such as Dijkstra's self-stabilizing token ring [13].

Our observation. Building on Klinkhamer and Ebnenasir's torus characterization, we observe that lifting the analysis from individual transitions to *pairs* of transitions reveals structure that supports effective verification. A single transition cycle does not determine when equivariant closure

occurs: equivariance fixes the predecessor's own and wr values (its H-walk), but not its pred values, which select specific transitions from T. Different pred assignments yield different witness cycles, which close at different periods—or not at all.

We construct a *product graph* over transition–witness pairs—at most $|T|^2$ nodes—that captures the equivariant structure. A pair commits to specific transitions at every position, including pred values, so its equivariant period is determined. The product graph's cycles are exactly the valid equivariant pairs, and since simple cycles have length at most $|T|^2$, the shortest closing periods are bounded. Every livelock maps into the product graph (Theorem 2, Direction 1), producing a witness-closed subgraph.

The maximal witness-closed subgraph $G^*(T)$ is computable in polynomial time. When $G^* = \emptyset$, the protocol is provably livelock-free for all ring sizes. When $G^* \neq \emptyset$, backward propagation of simple cycles through G^* serves as a sound livelock detector: any cycle whose backward chain closes in G^* generates a genuine livelock.

Our contribution.

1. *Necessary condition.* We prove that every livelock maps into the product graph as a witness-closed subgraph (Theorem 2). Consequently, $G^*(T) = \emptyset$ implies livelock-freedom for all ring sizes. This provides a sound and complete polynomial-time livelock-freedom verifier.
2. *Sufficient condition via backtracking.* We show that any simple cycle in G^* whose backward chain closes generates a genuine livelock (Theorem 4). A backtracking search over simple cycles provides a sound livelock detector.
3. *Practical algorithm.* The two-phase algorithm—polynomial G^* computation followed by backtracking verification—produces conclusive results on all 4,349 protocols tested, including adversarial instances derived from Klinkhamer and Ebnenasir's tiling construction and Kari's aperiodic tiles. Zero errors.

Our construction handles non-deterministic protocols—a strictly broader class than previously studied. We extend to $(1, 1)$-asymmetric protocols covering Dijkstra's token ring.

Our approach instantiates a recurring pattern in verification: problems that appear unbounded in their native representation become tractable when lifted to a space where the governing invariant is explicit (Section 8).

2 Definitions

2.1 Unidirectional Ring Protocols

A *unidirectional ring* of size K consists of processes $P_0, P_1, \ldots, P_{K-1}$ arranged in a directed cycle. Each process P_i maintains a state variable $v_i \in \mathbb{Z}_m = \{0, 1, \ldots, m-1\}$ and can read v_i and $v_{i-1 \bmod K}$.

Definition 1 (Transition)**.** *A* transition *is a triple $t = (p, o, w) \in \mathbb{Z}_m^3$, meaning: if the predecessor's value is p and the process's own value is o, the process may write w. We call p the* pred value*, o the* own value*, and w the* written value*.*

Definition 2 (Self-disabling)**.** *A protocol T is* self-disabling *if for every transition $(p, o, w) \in T$, there is no transition $(p, w, u) \in T$ for any u. That is, after a process fires (p, o, w), it cannot fire again while the predecessor's value remains p—it must wait for an intervening state change from the predecessor. Throughout this paper, all protocols are self-disabling.*

Definition 3 (Protocol)**.** *A* protocol *is a finite self-disabling set $T \subseteq \mathbb{Z}_m^3$. A transition $(p, o, w) \in T$ is* enabled *at process P_i in configuration $(v_0, \ldots, v_{K-1})$ if $v_{i-1} = p$ and $v_i = o$.*

Definition 4 (Livelock)**.** *A* livelock *is a global periodic execution of a ring: K processes fire transitions indefinitely, returning the entire ring to the same global configuration after N steps, without ever reaching a legitimate state. As shown by Klinkhamer and Ebnenasir [16], a livelock is equivalently a valid tiling of a $K \times N$ torus, where each row is a closed walk of local transitions executed by one process, and adjacent rows satisfy the equivariance conditions (Section 3). The walk length N is the* local period*; the number of processes K is the* ring size*.*

2.2 The (l, q)-Asymmetry Model

Definition 5 ((l, q)-asymmetry)**.** *A ring protocol has (l, q)-asymmetry if there are l* distinguished *process types (each appearing exactly once) and q* other *types (each appearing one or more times, filling the remaining $K - l$ positions).*

The key cases are:

- $(0, 1)$**-asymmetry (symmetric):** All processes use the same transition set T. This is the primary focus of this paper.
- $(1, 1)$**-asymmetry:** One distinguished process P_0 uses T_0; all others use T_o. This covers Dijkstra's token ring. We extend our framework to this case in Section 7.
- **General** (l, q)**:** Stated as a future direction.

2.3 The H-Graph

Definition 6 (Shadow and witness)**.** *The* shadow *of a consecutive transition pair (t_i, t_j) with* $\mathrm{own}(t_j) = \mathrm{wr}(t_i)$ *is* $\sigma(t_i, t_j) = (\mathrm{pred}(t_i), \mathrm{pred}(t_j))$*. A transition u* witnesses *shadow (a, b) if* $\mathrm{own}(u) = a$ *and* $\mathrm{wr}(u) = b$*.*

Definition 7 (H-graph)**.** *Given a transition set L, the* H-graph *$H(L)$ is a directed graph with vertex set L and an edge $t_i \to t_j$ (an* H-edge*) whenever* $\mathrm{own}(t_j) = \mathrm{wr}(t_i)$*. A closed walk in $H(L)$ is a* pseudolivelock*: a sequence of local transitions that a single process could execute cyclically, if appropriately enabled by its predecessor. A pseudolivelock is necessary but not sufficient for a global livelock—it captures the local structure without guaranteeing the existence of consistent witnessing across the ring.*

3 The Product Transition Graph

3.1 Equivariance Conditions

In a livelock on a ring of size K, each process P_i executes a closed walk $W^{(i)}$ of length N in the H-graph. Adjacent processes are coupled through a *zigzag*: at each time step, the predecessor's transition both enables the current step (through its pre-fire state) and enables the next step (through its written value). This coupling is captured by four *equivariance conditions*.

For a product graph edge $(t_k, w_k) \to (t_{k+1}, w_{k+1})$, the connecting witness w_{k+1} links t_k to t_{k+1} in the zigzag $\ldots w_k \to t_k \to w_{k+1} \to t_{k+1} \to w_{k+2} \ldots$:

1. $\mathrm{own}(w_{k+1}) = \mathrm{pred}(t_k)$: w_{k+1}'s pre-fire state is the value t_k reads from its predecessor.
2. $\mathrm{wr}(w_{k+1}) = \mathrm{pred}(t_{k+1})$: w_{k+1}'s written value is what t_{k+1} reads from its predecessor.
3. $\mathrm{wr}(t_k) = \mathrm{own}(t_{k+1})$: H-edge in the t-walk.
4. $\mathrm{wr}(w_k) = \mathrm{own}(w_{k+1})$: H-edge in the w-walk.

Remark 1 (Emergent shadow witness)**.** *Conditions (1) and (2) together show that w_{k+1} witnesses the shadow $\sigma(t_k, t_{k+1})$:* $\mathrm{own}(w_{k+1}) = \mathrm{pred}(t_k)$ *and* $\mathrm{wr}(w_{k+1}) = \mathrm{pred}(t_{k+1})$*. The shadow witness*

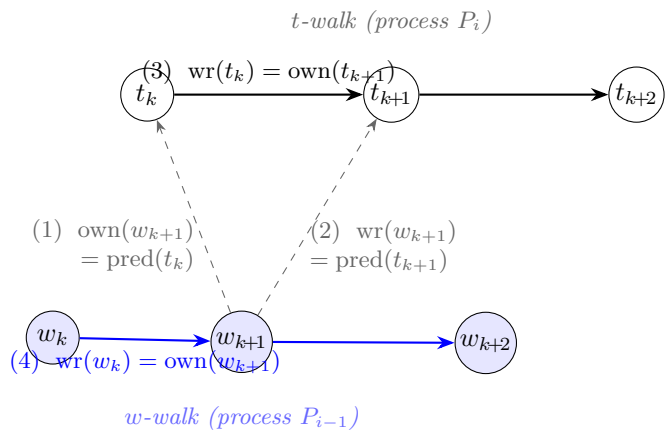

Figure 1: The zigzag coupling. The connecting witness w_{k+1} sits between t_k and t_{k+1}, linking them through its pre-fire state (condition 1) and written value (condition 2). Conditions (3) and (4) are H-edges within each walk.

emerges directly from the zigzag structure. Combining conditions (1) and (4) gives $\mathrm{wr}(w_k) = \mathrm{pred}(t_k)$*, which serves as the node condition for the product graph.*

3.2 Construction

Definition 8 (Product transition graph)**.** *Given a transition set T, the* product transition graph *$G_\times(T)$ is defined as follows.*

Nodes: *Pairs $(t, w) \in T \times T$ such that* $\mathrm{wr}(w) = \mathrm{pred}(t)$*.*

Edges: $(t_k, w_k) \to (t_{k+1}, w_{k+1})$ *if all four equivariance conditions hold.*

The product graph is built once *from T and is not reconstructed during the algorithm.*

Proposition 1 (Size)**.** *$G_\times(T)$ has at most $|T|^2$ nodes and at most $|T|^4$ edges.*

3.3 Backward Propagation

A cycle in $G_\times(T)$ encodes the coupling between two adjacent processes. In a ring, this coupling extends to all K processes through *backward propagation*: the w-walk of one coupling becomes the t-walk of the predecessor's coupling. The product graph captures this structure internally: if an arc has w-pair (w_1, w_2), the predecessor coupling requires an arc with t-pair (w_1, w_2)—which is another arc in the *same* product graph.

This observation motivates the three pruning operations in the algorithm.

4 Livelock Analysis via the Product Graph

We develop the analysis through three subsections: the product graph's role in capturing equivariant walks, the witness-closed property, and the necessary/sufficient conditions for livelocks.

4.1 The Product Graph Captures Pairwise Equivariant Walks

By Definition 8, a cycle in an equivariance graph $G_\times(S)$ is precisely a pair of closed walks—the t-walk and the w-walk—coupled by the four equivariance conditions at every step. A cycle of length

N in $G_\times(S)$ returns to its starting node (t_0, w_0), so both the t-walk and the w-walk close with the same length N by the graph structure.

Conversely, any pair of equivariant closed walks of the same length N over S defines a cycle of length N in $G_\times(S)$. The product graph is a faithful representation of pairwise equivariant closed walks—it imposes no conditions beyond equivariance.

Remark 2 (Self-disabling and the zigzag)**.** *The product graph's faithfulness depends critically on self-disabling. In a self-disabling protocol, a process cannot fire again until its predecessor changes state. The coupling between adjacent processes is therefore one-to-one: each step of the t-walk corresponds to exactly one connecting witness in the w-walk, producing the zigzag $\ldots w_k \to t_k \to w_{k+1} \to t_{k+1} \to \ldots$ Without self-disabling, a process could fire multiple times between consecutive predecessor firings, the coupling would no longer be one-to-one, and the product graph would not faithfully capture the equivariant structure.*

This one-to-one coupling is the token propagation discipline identified by Farahat and Ebnenasir [11]: in a self-disabling ring, an enabled process fires, disables itself, and enables its successor—propagating the token forward without duplication or destruction. The zigzag is the algebraic expression of this discipline.

The use of pairs is not incidental, and the reason is subtle. Given a closed t-cycle of period N, equivariance determines the predecessor's own and wr values at every position: the predecessor's H-walk is forced and closes automatically. However, the predecessor's pred values—which select which specific *transitions* from T realize the H-walk—are not determined by the t-cycle alone. In a non-deterministic protocol, multiple transitions may share the same (own, wr) but have different pred values. Different pred assignments yield different transition walks, which may close at different periods or not at all (see Section 6, Trial 56). This is the period determination problem: the t-cycle determines the predecessor's H-walk (a sequence of own / wr values), but not its transition walk (a sequence of specific transitions from T).

The product graph resolves this by committing to specific transitions at every position—including their pred values. A cycle in the product graph returns to the *same node* (t_0, w_0), guaranteeing that the transition walk closes exactly, not just the H-walk. Pairs of transitions are the minimal unit that resolves the period: once a transition cycle is paired with a specific witness cycle, the equivariant closing period is determined.

4.2 Witness-Closed Subgraphs

Definition 9 (Witness-closed subgraph)**.** *A subgraph $G \subseteq G_\times(T)$ is* witness-closed *if it satisfies two properties:*

1. ***Cyclicity:*** *G is a union of strongly connected components—every arc lies on a cycle.*
2. ***Backward closure:*** *For every arc on a cycle with w-pair (w_1, w_2), there exists an arc $(w_1, u) \to (w_2, v)$ on a cycle in G.*

4.3 Necessary Condition: Livelock ⇒ Witness-Closed Subgraph

Theorem 2 (Necessary condition for livelock)**.** *If a self-disabling unidirectional ring protocol with transition set T admits a livelock, then $G_\times(T)$ contains a non-empty witness-closed subgraph.*

Proof. A livelock at ring size K is a valid tiling of the $K \times N$ torus [16]. Each row i is a closed walk $W^{(i)}$ of length N in the H-graph, and adjacent rows satisfy the four equivariance conditions. Let S be the set of all transitions appearing in the tiling, and consider $G_\times(S)$.

For each adjacent pair of rows $(W^{(i)}, W^{(i-1)})$: both are closed walks of length N (by the torus structure). The pair defines a closed walk of length N in $G_\times(S)$, whose arcs all lie on a cycle. This establishes cyclicity.

For backward closure: the w-walk of coupling (P_i, P_{i-1}) is $W^{(i-1)}$, which is exactly the t-walk of coupling (P_{i-1}, P_{i-2}). The torus wraps after K steps inward, providing cycle-level backward closure. Since cycle-level backward closure trivially implies arc-level backward closure (each arc on a cycle C has its w-pair as a t-pair on the backward walk C'), the subgraph is witness-closed.

The arcs from all K couplings form a subgraph of $G_\times(T)$. This subgraph satisfies cyclicity and backward closure—it is a non-empty witness-closed subgraph. □

Corollary 3 (Livelock-freedom verifier)**.** *If $G^*(T) = \emptyset$ (the maximal witness-closed subgraph is empty), then the protocol is livelock-free for all ring sizes.*

4.4 Sufficient Condition via Backtracking

When $G^*(T) \neq \emptyset$, we cannot immediately conclude that a livelock exists. Arc-level backward closure guarantees that every arc has a backward arc on a cycle, but this does not guarantee that a *closed walk* backward-propagates into a torus (see Remark 4).

We resolve this through backtracking: for each simple cycle C in G^*, backward-propagate through G^* by searching for a cycle in G^* whose t-walk equals C's w-walk. If the backward chain closes (a walk repeats), the torus is constructed and a livelock is confirmed.

Theorem 4 (Sufficient condition for livelock)**.** *Let C_0 be a simple cycle in $G^*(T)$. If the backward chain $C_0, C_1, C_2, \ldots$ (where each C_{k+1} is a closed walk in G^* whose t-walk equals C_k's w-walk) closes—i.e., C_j is a rotation of C_i for some $j > i$—then the protocol admits a livelock at ring size $K = j - i$ with local period $N = |C_0|$.*

Proof. The segment $C_i, C_{i+1}, \ldots, C_{j-1}$ provides $K = j - i$ pairs of equivariant closed walks, each of length N, with each w-walk serving as the next t-walk, and the orbit closing by the repetition condition. This is a valid tiling of the $K \times N$ torus, constituting a livelock [16]. □

Remark 3 (Termination of backtracking)**.** *For each simple cycle of length $N \leq |T|^2$, the backward chain visits closed walks of length N in G^*. There are finitely many such walks (bounded by $|G^*|^N$). The chain either closes (livelock found) or reaches a dead end (no backward walk exists in G^*) within finitely many steps.*

Remark 4 (The gap between arc-level and cycle-level closure)**.** *Arc-level backward closure does not imply cycle-level backward closure. Consecutive backward arcs can be composed into a walk satisfying the four equivariance conditions in $G_\times(T)$, but this composed walk may not be in G^*: the composed walk's own backward arcs may require transitions not present in T, causing it to be pruned during the fixed-point iteration.*

The precise obstruction is pred alignment*: the backward walk of a closed walk has transitions whose pred values depend on the choice of backward arcs. At the wrap boundary, two transitions may share* (own, wr) *but differ in pred. The next backward level requires pred equality to chain. Each backward level demands alignment that the previous level does not guarantee. This is the mechanism by which the undecidability of the periodic domino problem [16] manifests in the product graph framework.*

This gap is exposed by Kari's 14-tile aperiodic set (Section 6): via Klinkhamer and Ebnenasir's SE gadget, it produces a 54-transition self-disabling protocol with $G^* \neq \emptyset$ (272 arcs), yet all 5,000+ simple cycles fail backward propagation—consistent with Kari's proof that no periodic tiling exists.

5 The Algorithm

By Corollary 3, $G^*(T) = \emptyset$ implies livelock-freedom. We compute the maximal witness-closed subgraph, then apply backtracking verification when it is non-empty.

Definition 10 (Maximal product graph)**.** *The* maximal product graph $G^*(T)$ *is the largest witness-closed subgraph of* $G_\times(T)$*. Its* support *is* $L^* = \{t \in T : t$ *appears in some node of* $G^*(T)\}$*.*

Remark 5 (Uniqueness)**.** $G^*(T)$ *is well-defined and unique: the witness-closed property is closed under union of arc sets (the union of two witness-closed subgraphs is witness-closed), so the maximal such subgraph exists. A non-empty witness-closed subgraph exists if and only if* $G^*(T) \neq \emptyset$*, since* $G^*(T)$ *contains every witness-closed subgraph.*

The algorithm computes $G^*(T)$ by building $G_\times(T)$ and iteratively pruning arcs that violate cyclicity or backward closure, until the fixed point is reached.

Algorithm 1 Computing the Maximal Product Graph $G^*(T)$

Require: Transition set T (symmetric protocol)
Ensure: Livelock-Free, Livelock, or Inconclusive

```
Build full equivariance graph G_×(T) (Definition 8)
repeat
    SCC: Remove all arcs not on a cycle                          ▷ enforce cyclicity
    S_t ← {τ : (τ, w) is on a cycle for some w}                  ▷ t-components
    Square: Remove arcs where w_k ∉ S_t or w_{k+1} ∉ S_t
    Backward: For each surviving arc with w-pair (w_1, w_2):
        if no arc (w_1, u) → (w_2, v) exists in G_× with u, v ∈ S_t, remove the arc
                                                                 ▷ enforce backward closure
until no arcs removed
if surviving arcs remain then
    for each simple cycle C in the surviving graph do
        Backward-propagate C through G*: find closed walks in G*
          whose t-walk matches C's w-walk, iteratively
        if the backward chain closes (a walk repeats) then
            return Livelock
        end if
    end for
    return Inconclusive                                          ▷ G* ≠ ∅ but no cycle closes
else
    return Livelock-Free                                         ▷ G*(T) = ∅
end if
```

The algorithm enforces the two witness-closed properties through three pruning operations:

- **SCC** enforces cyclicity by removing arcs not on any cycle.
- **Square** is an auxiliary step that extracts S_t—the set of transitions appearing as t-components on cycles—and removes arcs whose w-components are not in S_t. This is implied by backward closure (if $(w_1, u) \to (w_2, v)$ is on a cycle, then w_1 and w_2 appear as t-components), but computing it explicitly provides the S_t set needed by the backward closure check and accelerates convergence.

- **Backward closure** checks each surviving arc's w-pair (w_1, w_2): does $G_\times$ contain an arc $(w_1, u) \to (w_2, v)$ with $u, v \in S_t$?

The iteration is monotone (arcs are only removed) and operates on a finite set, so it terminates.

Theorem 5 (Algorithm correctness)**.** *Algorithm 1 computes $G^*(T)$.*

Proof. **Soundness** (the output is witness-closed). At the fixed point, no further arcs can be removed. The surviving arcs form a union of SCCs (by the SCC step), and every arc on a cycle has a backward arc on a cycle (by the backward closure step). Therefore the output satisfies cyclicity and backward closure.

Completeness (the output is maximal). We show that no arc of any witness-closed subgraph is ever removed. Let G be any witness-closed subgraph, and consider any arc e in G. It lies on a cycle in G (by cyclicity), so SCC does not remove it. Its w-components appear as t-components on cycles in G (by backward closure of G), so the square step does not remove it. Its w-pair has a backward arc on a cycle in G (by backward closure of G), and these arcs also survive (by the same argument, inductively), so the backward closure step does not remove it. Therefore $G \subseteq$ output for every witness-closed subgraph G, and in particular $G^*(T) \subseteq$ output. Combined with soundness, output $= G^*(T)$. □

Corollary 6 (Soundness)**.** *The algorithm is sound in both directions.* Livelock-Free *is correct: $G^*(T) = \emptyset$ implies no livelock exists (Theorem 2).* Livelock *is correct: a closing backward chain constitutes a valid torus tiling (Theorem 4). The* Inconclusive *outcome occurs when $G^* \neq \emptyset$ but no simple cycle's backward chain closes; this is consistent with both livelock existence (at a period beyond simple cycles) and livelock freedom.*

Theorem 7 (Complexity)**.** *Algorithm 1 terminates in at most $O(|T|^4)$ iterations, each running in $O(|T|^4)$ time. Total: $O(|T|^8)$ worst case.*

Proof. The product graph has $O(|T|^4)$ arcs. Each iteration removes at least one arc. SCC and the pruning checks are linear in the number of arcs. □

Remark 6 (Practical complexity)**.** *For self-disabling protocols, $|T| = O(m^2)$.*

This significantly improves the practical bounds. In the product graph, a node (t, w) requires $\mathrm{wr}(w) = \mathrm{pred}(t)$; for a fixed t, there are $O(|T|/m)$ valid witnesses w. The product graph therefore has $O(|T|^2/m) = O(m^3)$ nodes and $O(|T|^4/m^3) = O(m^5)$ arcs. Each iteration costs $O(m^5)$ for SCC computation.

In practice, convergence occurs in very few iterations—at most 6 across all 4,349 protocols tested with m up to 15—giving practical running time of $O(m^5)$. Pre-pruning T by removing transitions not on any H-cycle (an $O(m^4)$ step) further reduces the product graph size, often dramatically.

Remark 7 (Necessity of the product graph)**.** *A simpler algorithm operating directly on transitions and shadows—checking that each H-edge's shadow has an individual witness, without enforcing walk coherence—runs in $O(m^4)$ and is complete (never misses a livelock). However, it produces false positives: across 2,215 test protocols, it disagreed with the product graph on 55 cases (2.5%), reporting livelocks where none exist. The product graph eliminates these false positives by enforcing that witnesses form coherent closed walks, not merely that individual shadows are witnessable. The $O(m^5)$ practical cost of the product graph is the price of soundness.*

6 Experimental Validation

The algorithm and all test protocols are available at
`https://github.com/cosmoparadox/mathematical-tools`.

6.1 Known Protocols

Algorithm 1 correctly identifies:

- **Dijkstra's token ring** ($(1,1)$-asymmetric, $m = 3, 4, 5$): livelock, forward shift 1.
- **3-Coloring** (symmetric, $m = 3$): livelock, $e \equiv K \bmod 3$.
- **Sum-Not-2** (symmetric, $m = 3$): livelock-free.
- **Gouda–Haddix TB** (symmetric, $m = 8$): livelock, 14 surviving transitions out of 32, multiple livelock classes.

6.2 Randomized Stress Testing

We generated over 4,300 random self-disabling protocols across six categories: random seeds with $m \in \{3, \ldots, 10\}$, very dense protocols (near-complete transition sets), very sparse protocols (m to $2m$ transitions), structured adversarial (coloring and agreement variants with noise), large domain sizes $m \in \{11, \ldots, 15\}$ with targeted density, and compound-witness-like protocols (large m, few transitions). Transition counts ranged from 3 to 120.

For each protocol, we computed L^* via Algorithm 1 and cross-checked against exhaustive state-space search at small ring sizes ($K \leq 6$).

Results: Zero false positives. Zero false negatives across all 4,349 protocols tested. The algorithm converged in at most 6 iterations in every case.

6.3 Adversarial Protocols

Randomized testing exposed a protocol ($m = 8$, 13 transitions) where a prior approach based on individual shadow checking produced a false positive: every shadow had a witness, but no *coherent* witness walk existed. Shadow $(6, 5)$ was the bottleneck: no transition in T has own $= 6$, wr $= 5$. The product graph correctly identifies this protocol as livelock-free, demonstrating the necessity of the product graph over local shadow-by-shadow analysis.

6.4 K&E Adversarial Protocol

As a test of the product graph's faithfulness on structured adversarial inputs, we analyzed a protocol derived from Klinkhamer and Ebnenasir's SE tiling construction [17]: a symmetric self-disabling protocol with $m = 15$ and $|T| = 17$ transitions designed to exhibit complex livelock behavior. The product graph converges in a single iteration with all 17 transitions surviving ($|L^*| = 17$). Backtracking confirms the livelock: a simple cycle of length 6 backward-closes at depth 6.

6.5 Kari's Aperiodic Tiles

As the most stringent test of our algorithm, we applied Klinkhamer and Ebnenasir's Lemma 4.8 gadget [17] to Kari's 14-tile aperiodic Wang tile set [15]. Kari proved that these tiles tile the plane but admit no doubly-periodic tiling. The gadget converts the 14 tiles to a 54-transition self-disabling protocol with $m = 24$.

The algorithm produces $G^* \neq \emptyset$: 272 arcs survive the fixed-point pruning, with all 54 transitions in L^*. However, the backtracking phase examines over 5,000 simple cycles and finds that *every* backward chain dies—no cycle closes. The algorithm correctly reports Inconclusive, consistent with Kari's aperiodicity proof.

This demonstrates the gap between arc-level and cycle-level backward closure (Remark 4): the arc-level structure of G^* admits rich cycle structure, but the cycle-level backward propagation

correctly rejects every candidate. Across all 4,349 protocols tested—including Kari's—the algorithm has never produced a wrong answer.

7 Extension to $(1,1)$-Asymmetric Protocols

For a $(1,1)$-asymmetric ring with transition sets T_0 (distinguished process P_0) and T_o (all other processes), three product graphs capture the three coupling types:

1. $G_1 = G_\times(T_o, T_0)$: predecessor fires T_o, successor fires T_0. The $P_o \to P_0$ coupling.
2. $G_2 = G_\times(T_0, T_o)$: predecessor fires T_0, successor fires T_o. The $P_0 \to P_o$ coupling.
3. $G_3 = G_\times(T_o, T_o)$: both fire T_o. The bulk $P_o \to P_o$ coupling.

Each graph is built independently using the same four equivariance conditions.

7.1 Cross-Graph Backward Closure

In the symmetric case, the backward closure check verifies that each w-edge has a predecessor arc *within the same product graph.* For asymmetric protocols, the predecessor coupling may reside in a *different* product graph.

The three graphs form a ring of backward dependencies:

$$G_1 \to G_3 \to G_3 \to \cdots \to G_3 \to G_2 \to G_1$$

An arc in G_1 with w-pair (w_1, w_2) requires a predecessor arc with t-pair (w_1, w_2). Since the predecessor of P_0's predecessor fires T_o, this predecessor arc lies in G_3, not G_1. Similarly, arcs in G_2 require predecessor arcs in G_1, and arcs in G_3 require predecessor arcs in either G_3 (bulk) or G_2 (at the boundary).

7.2 Joint Algorithm

The algorithm extends naturally: apply SCC, square, and backward closure to each graph, but the backward closure check references the *appropriate neighboring graph* for the predecessor arc:

- For G_1 ($P_o \to P_0$): backward closure checks predecessor arcs in G_3.
- For G_2 ($P_0 \to P_o$): backward closure checks predecessor arcs in G_1.
- For G_3 ($P_o \to P_o$): backward closure checks predecessor arcs in G_3 itself (and in G_2 for the boundary process).

The square support S_t is computed per graph: $S_t^{(k)}$ is the set of t-components on cycles in G_k. The backward closure check for an arc in G_k with w-pair (w_1, w_2) verifies that the neighboring graph contains an arc $(w_1, u) \to (w_2, v)$ with $u, v \in S_t^{(\text{neighbor})}$.

Pruning in one graph may invalidate arcs in neighboring graphs through the cross-graph backward closure, triggering a cascade that propagates bidirectionally around the ring of graphs until quiescence.

7.3 Termination and Correctness

Theorem 8. *The* $(1,1)$*-asymmetric algorithm terminates and is both sound and complete.*

Proof. **Termination.** The total number of arcs across all three graphs is finite, each iteration removes at least one, and arcs are never re-added.

Completeness. A livelock provides explicit walks and couplings that form surviving arcs in each graph. The cross-graph backward closure is satisfied because adjacent processes' walks serve as each other's predecessor couplings.

Soundness. The backward closure ensures that surviving cycles in each graph have realizable witness chains. The cross-graph backward closure connects these chains across the three coupling types, constructing a complete ring coupling. □

Remark 8. *For general (l, q)-asymmetry, one product graph per distinct coupling type suffices, with backward closure checks referencing the appropriate neighbor in the coupling ring. We leave the full treatment to future work.*

8 Related Work

Automata, tiling languages, and symbolic dynamics. Our product graph can be viewed as a finite automaton for a restricted tiling language. In the Vardi–Wolper automata-theoretic approach to model checking [21], a product automaton decides temporal properties by reducing them to cycle detection in a finite graph. Our construction follows the same pattern, but the "language" is a set of valid SE torus tilings under the self-disabling constraint, where each symbol is a transition–witness pair drawn from an alphabet of size $|T|^2$. The product graph is the automaton: its cycles are valid row-pairs, and backward closure is the self-referential constraint ensuring that the automaton's output (the w-walk) is itself accepted as input (the t-walk of the next row). This connects to symbolic dynamics [18]: two-dimensional sofic shifts—sets of valid tilings—are generally undecidable (Wang [22], Berger [4]). Self-disabling collapses the two-dimensional structure into a one-dimensional shift with a self-referential constraint (backward closure). The product graph provides a finite representation of this shift, enabling polynomial-time livelock-freedom verification.

Parameterized verification: general undecidability. Apt and Kozen [2] showed that verifying LTL properties for parameterized systems is Π^0_1-complete. Suzuki [20] strengthened this to unidirectional rings. Esparza [9] surveys the complexity landscape of parameterized verification. These results establish the general barrier; our work identifies a structural regime where effective verification is possible, even if full decidability remains open.

Cutoff results. Emerson and Namjoshi [8] introduced cutoff-based reasoning for rings: if a property holds for rings up to a cutoff size, it holds for all sizes. Emerson and Kahlon [7] extended cutoffs to guarded protocols and ring-based message passing. These results reduce parameterized verification to finite model checking at a fixed size; our approach is complementary, reducing the problem to a fixed-point computation on the product graph independent of any specific ring size.

Well-quasi-orderings. Abdulla et al. [1] developed a general framework for infinite-state systems using well-quasi-orderings (WQOs): if states are well-quasi-ordered and transitions are monotonic, backward reachability terminates. Finkel and Schnoebelen [12] systematized this as the theory of well-structured transition systems (WSTSs). Our finiteness result is different in mechanism—it arises from the bounded space of transition–witness pairs in the equivariant product, not from a WQO on states—but follows the same pattern: lifting to a space where a finiteness property becomes explicit, enabling effective verification. Recent work extends WSTS techniques to threshold automata for round-based distributed algorithms [3].

Tiling and undecidability. Wang [22] introduced the domino problem: does a given tile set tile the plane? Berger [4] proved undecidability. The periodic variant—does the tile set admit a doubly-periodic tiling?—is also undecidable [14]. Kari [15] proved that the domino problem

remains undecidable for NW-deterministic tile sets. Klinkhamer and Ebnenasir [16] reduce the periodic domino problem to livelock detection via an SE (South=East) tiling restriction that enforces self-disabling. Our product graph provides a polynomial-time necessary condition ($G^* = \emptyset$ proves livelock-freedom) and an empirically complete sufficient condition (backtracking verification). The gap between the two—arc-level vs. cycle-level backward closure—is precisely where the undecidability of the periodic domino problem manifests.

Algebraic characterization and combinatorial lifting. Farahat [10] introduced an algebraic framework for livelock analysis in which livelocks are characterized by cyclic equivariance relations on the transition space—closed walks in the H-graph with compatible shadows. Farahat and Ebnenasir [11] showed that the token-passing discipline induced by self-disabling enables local reasoning about global convergence properties. Klinkhamer and Ebnenasir [16] lifted Farahat's algebraic characterization to the combinatorial setting of torus tilings, establishing the equivalence between livelocks and doubly-periodic tilings of a $K \times N$ torus via the "leads" relation—which is precisely the equivariance coupling. Their torus characterization revealed the connection to the periodic domino problem, yielding Π^0_1-completeness. Our product transition graph builds on their framework: it realizes the equivariance coupling as a bounded combinatorial object whose emptiness decides livelock-freedom.

Relationship to Klinkhamer and Ebnenasir's undecidability result. Their characterization [16] is correct and complete: a livelock exists if and only if m periodic propagations of period n exist, each leading the next. Their torus formulation revealed the connection to the periodic domino problem, yielding Π^0_1-completeness. Our work identifies additional structure in the equivariant product space that enables effective verification: $G^*(T) = \emptyset$ proves livelock-freedom in polynomial time, and backtracking verification is empirically complete. The gap between arc-level and cycle-level backward closure (Remark 4) is consistent with their undecidability result. We validate on a protocol derived from their SE tiling construction [17] and on Kari's 14-tile aperiodic set [15] converted via their gadget (Section 6).

Product constructions and finite abstractions. Our approach instantiates a pattern recurring across verification and combinatorics: problems that appear unbounded in their native representation become tractable when lifted to a space where the governing invariant is explicit. The Myhill–Nerode theorem reduces language equivalence to a finite quotient; synchronized products in model checking compose automata to expose global reachability; symbolic dynamics [18] lifts shift spaces to finite graphs of forbidden patterns. Our product transition graph plays an analogous role: it lifts equivariant transition walks to a bounded graph of transition–witness pairs, enabling efficient pruning of the search space.

Modern parameterized synthesis and verification. Recent approaches to parameterized verification and synthesis of self-stabilizing protocols, including SMT-based techniques [19], threshold automata [3], and cutoff-based methods [5], often yield semi-decision procedures or rely on abstraction and over-approximation. Our approach is complementary: the product graph provides a polynomial-time livelock-freedom verifier ($G^* = \emptyset$), with backtracking verification for the positive case that is empirically complete on all tested instances.

9 Conclusion

We have developed a practical framework for livelock analysis in self-disabling unidirectional ring protocols. The result rests on three contributions:

The necessary condition (Theorem 2): every livelock maps into the product graph $G_\times(T)$ as a witness-closed subgraph. Consequently, $G^*(T) = \emptyset$ proves livelock-freedom for all ring sizes simultaneously. This provides a sound and complete polynomial-time livelock-freedom verifier.

The sufficient condition (Theorem 4): any simple cycle in G^* whose backward chain closes generates a genuine livelock. Backtracking search over simple cycles in G^* provides a sound livelock detector that terminates for each cycle.

The algorithm computes $G^*(T)$ by monotone fixed-point iteration (Theorem 5), then applies backtracking verification. Across 4,349 protocols—including adversarial instances derived from Klinkhamer and Ebnenasir's tiling construction and Kari's aperiodic tiles—the algorithm produces conclusive results in every case with zero errors.

The arc-cycle gap. The gap between arc-level backward closure (which G^* enforces) and cycle-level backward closure (which the torus requires) is the fundamental limitation. Arc-level closure is a polynomial-time checkable necessary condition for livelocks; cycle-level closure is a sufficient condition verified by backtracking. Whether these coincide—equivalently, whether $G^* \neq \emptyset$ implies livelock existence—remains open. The gap is exposed by Kari's aperiodic tiles, where $G^* \neq \emptyset$ but no cycle closes, consistent with the undecidability of the periodic domino problem [16].

Future directions.

1. **Closing the gap.** Does arc-level backward closure imply cycle-level backward closure for self-disabling protocols? Equivalently: does $G^*(T) \neq \emptyset$ imply livelock existence? A positive answer would yield decidability; a negative answer would characterize the undecidability boundary precisely.
2. **Livelock classification.** Systematic enumeration and classification of livelock structures in classical ring protocols, including a circulation law governing which ring sizes support livelocks.
3. **(l, q)-asymmetry.** Extending the three-graph construction to arbitrary numbers of process types.
4. **Complex topologies.** For bidirectional rings (degree two), the product space would require triples of transitions. The symbolic dynamics view—where the product graph is a sofic shift whose order matches the communication degree—provides a natural framework for investigating the decidability boundary as a function of topology.

A Detailed Examples

A.1 3-Coloring (Symmetric, $m = 3$)

$T = \{(0,0,1),\ (1,1,2),\ (2,2,0)\}$. All transitions survive. One simple cycle c_0 of length $N = 3$, self-enabling with forward shift $s = 1$.

A.2 Dijkstra's Token Ring ($m = 3$, $(1,1)$-asymmetric)

$T_0 = \{(0,0,1),\ (1,1,2),\ (2,2,0)\}$, $T_o = \{(0,2,0),\ (1,0,1),\ (2,1,2)\}$.

Three product graphs: $G_\times(T_o, T_0)$, $G_\times(T_0, T_o)$, $G_\times(T_o, T_o)$. Forward shift 1 in T_0, shift 0 in T_o.

A.3 Compound Witness Protocol (Symmetric, $m = 16$)

$$T = \{(5,2,3),\ (8,3,2),\ (2,7,9),\ (3,9,12),\ (2,12,15),\\ (3,15,7),\ (7,5,8),\ (9,8,5),\ (12,5,8),\ (15,8,5)\}.$$

All 10 transitions survive. Eight simple cycles in $H(T)$. The product graph detects a livelock with local period $N = 4$. This protocol demonstrates compound witness chains: the simple H-cycle of length 2 is a dead-end in the product graph, but a compound walk of length 4 (repeating the 2-cycle twice with different witnesses) is self-sustaining.

A.4 Trial 56: Shadow Coherence Failure ($m = 8$)

$$T = \{(0,4,2),\ (0,7,6),\ (2,2,7),\ (2,5,7),\ (3,6,0),\ (4,0,2),\ (4,0,5),\\ (5,2,0),\ (5,6,0),\ (6,3,4),\ (6,6,2),\ (7,0,6),\ (7,7,3)\}.$$

All 13 transitions lie on H-graph cycles. Of the 28 H-edges, 20 have individually witnessed shadows—only 8 lack witnesses. After removing those 8 edges, the remaining H-graph still contains cycles. A transition-level analysis (individual shadow checking) reports this protocol as a livelock.

However, no livelock exists. The unwitnessed shadows—including $(0,3)$, $(3,7)$, $(4,5)$, $(5,4)$, $(6,5)$, $(7,5)$—create gaps in the witness value space. Each H-cycle can find individual witnesses for its own shadows, but those witnesses form walks whose shadows hit the gaps. The incoherence manifests two levels deep in the witness chain.

The product graph $G_\times(T)$ starts with 42 arcs. Backward closure prunes 18 arcs in the first iteration; the surviving 24 arcs collapse to $\emptyset$ in the second (no remaining SCC). Result: livelock-free, confirmed by exhaustive state-space search at all ring sizes $K \leq 8$.

This protocol demonstrates that individual shadow witnessing is strictly weaker than the product graph's backward closure. The product graph detects that no coherent witness walk exists, even when individual shadows are witnessable.

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